

LESSON 5-8 • SESSION 1 • PREREQUISITE PRACTICE B • SKILL 1 OF 2

Area of Triangles

Builds on Lesson 5-2 Session 1 • 6.G.A.1

TODAY'S LESSONYou will find one slanted face, like $\frac{1}{2} \times 12 \times 9$ for the trophy top.**YOU NEED TO**Find a triangle's area with $A = \frac{1}{2} \times b \times h$, like $\frac{1}{2} \times 14 \times 9$. **You will use it in Practice Form A, problems 1, 3, and 4.****STEPS TO FOLLOW***use these same steps on every problem*

- 1** Write $A = \frac{1}{2} \times b \times h$ with the numbers.
- 2** Halve the even number first.
- 3** Multiply what is left. Write square units.

WATCH ME*the steps in action***A face has a base of 14 cm and a height of 9 cm. Find its area.****Step 1** $A = \frac{1}{2} \times 14 \times 9$.**Step 2** Half of 14 is 7.**Step 3** $7 \times 9 = 63$, so 63 cm².**Answer** 63 cm²**TRY IT WITH HELP***fill in each step*

1. Base 10 in, height 17 in. Find the area.

Step 1 $A = \frac{1}{2} \times \underline{\hspace{2cm}} \times \underline{\hspace{2cm}}$ **Step 2** Half of 10 is $\underline{\hspace{2cm}}$ **Step 3** $5 \times 17 = \underline{\hspace{2cm}}$ in²

2. Base 6 m, height 11 m. Find the area.

Step 1 $A = \frac{1}{2} \times \underline{\hspace{2cm}} \times \underline{\hspace{2cm}}$ **Step 2** Half of 6 is $\underline{\hspace{2cm}}$ **Step 3** $3 \times 11 = \underline{\hspace{2cm}}$ m²**YOUR TURN***use the steps on your own*

3. Find each area. Use $A = \frac{1}{2} \times b \times h$.

a. $\frac{1}{2} \times 18 \times 11 = \underline{\hspace{2cm}}$

b. $\frac{1}{2} \times 20 \times 23 = \underline{\hspace{2cm}}$

c. $\frac{1}{2} \times 26 \times 15 = \underline{\hspace{2cm}}$

d. $\frac{1}{2} \times 8 \times 13 = \underline{\hspace{2cm}}$

e. $\frac{1}{2} \times 28 \times 17 = \underline{\hspace{2cm}}$

f. $\frac{1}{2} \times 24 \times 19 = \underline{\hspace{2cm}}$

Tell Slant Height From Height

Builds on Lesson 5-6 Session 1 • 6.G.A.4

TODAY'S LESSON

You will find each face with its slant height, like the 26 ft label on the roof cap.

YOU NEED TO

Know which length is the slant height: it runs down the middle of a face.

You will use it in Practice Form A, problems 2 and 5.

STEPS TO FOLLOW*use these same steps on every problem*

- 1 Find** the base edge of the face.
- 2 Pick** the slant height: it runs down the middle of a face.
- 3 Use** $\frac{1}{2} \times \text{base edge} \times \text{slant height}$.

WATCH ME*the steps in action*

A square pyramid has base edge 10 ft, height 12 ft, and slant height 13 ft. Find one face.

Step 1 Base edge: 10 ft.

Step 2 Slant height: 13 ft. Skip the 12 ft height.

Step 3 $\frac{1}{2} \times 10 \times 13 = 65$.

Answer **65 ft²**

TRY IT WITH HELP*fill in each step*

4. Base edge 16 cm, height 15 cm, slant height 17 cm. Find one face.

Step 1 Base edge: _____ cm

Step 2 Slant height: _____ cm

Step 3 $\frac{1}{2} \times 16 \times 17 =$ _____ cm²

5. Base edge 12 in, height 8 in, slant height 10 in. Find one face.

Step 1 Base edge: _____ in

Step 2 Slant height: _____ in

Step 3 $\frac{1}{2} \times 12 \times 10 =$ _____ in²

YOUR TURN*use the steps on your own*

6. Find one face. Each lists base edge, height, slant height.

a. 6, 4, 5 _____

b. 24, 5, 13 _____

c. 18, 12, 15 _____

d. 14, 24, 25 _____

READY CHECK*a problem like the one in today's lesson*

7. Base edge 8 m, height 3 m, slant height 5 m. What is the area of one face?

A 12 m²

B 20 m²

C 24 m²

D 40 m²

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #_____